

# LinkForge — Developer, GitHub & Deployment Runbook

Complete Engineering Guide, Remote VPS Deployment, Zero-Terminal Updates, Dad's Setup & Extension Signing (v2.4.2)

## 1. System Architecture & Components Overview

LinkForge is an autonomous personal intelligence hub and knowledge repository comprising six core micro-engines:

| Component           | File / Service                            | Key Responsibilities                                                                                               |
|---------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Web App & API       | app.py, blueprints/                       | Flask web framework, ProxyFix headers for SSL reverse proxies, REST APIs for browser extension, PWA share targets. |
| Neural Link Engine  | services/scraper.py                       | Full-article extraction, offline readability HTML & WebP packaging, auto-tagging, SQLite WAL database storage.     |
| Video & Transcripts | services/videos.py                        | YouTube transcript ingestion, timestamped navigation, auto-suggest categorization, video bookmarking.              |
| Book Vault Engine   | services/books.py                         | Autonomous Shelfmark grabbing, Hardcover.app API integration, EPUB streaming, keyword filtering & sorting.         |
| LinkForge Pulse     | services/pulse.py                         | AI interest cluster synthesis, multi-source RSS intelligence feeds, real-time news aggregations.                   |
| Updater & Backups   | services/updater.py<br>services/backup.py | Zero-terminal 1-click GitHub live updates, atomic SQLite snapshots, 12-table CSV exports, full archive packaging.  |

## 2. Project Directory Structure

|                       |                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| /blueprints/          | Modular routes: dashboard.py (home, books, pulse), links.py, videos.py, admin.py, notes.py                      |
| /services/            | Core background engines: books.py, pulse.py, backup.py, nuke.py, scraper.py, updater.py, db.py                  |
| /templates/           | Jinja2 UI templates: books.html, pulse.html, admin_nuke.html, admin_extensions.html, base.html                  |
| /data/                | Persistent volume: linkforge.db (SQLite), /archives/ (offline web cache), /books/ (downloaded EPUBs), /backups/ |
| /dashforge-extension/ | Universal Chrome/Edge/Firefox WebExtension source code and manifest.                                            |
| docker-compose.yml    | Container definition, port mapping (5006:5000), volume bindings, healthcheck, and auto-restart policy.          |

### 3. How to Update Your GitHub Repository

Whenever you or Antigravity complete code changes on your local PC, push them to GitHub with this standard 3-command workflow:

```
# 1. Navigate to the project folder
cd X:\.config\docker\linkforge
# 2. Stage all modified files and commit with a description
git add .
git commit -m "Describe what you changed here (e.g. Add dark mode toggle)"
# 3. Push to your GitHub repository
git push origin main
```

**100% Privacy Guarantee:** The .gitignore configuration strictly excludes your ./data/ folder, SQLite databases (\*.db), downloaded EPUBs, cached article archives, and private secrets. Pushing to GitHub will **NEVER** upload your bookmarks, personal notes, or login credentials.

### 4. Setting Up Dad's Computer / Remote Devices (Step-by-Step)

When visiting your dad or setting up a family member's device, follow these quick steps to give them full access in under 5 minutes:

|                                       |                                                                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1: Open LinkForge</b>         | Open any browser on his PC, tablet, or phone and visit <a href="https://linkforge.okapitek.uk">https://linkforge.okapitek.uk</a> (or his Tailscale IP <a href="http://100.64.194.104:5006">http://100.64.194.104:5006</a> ).                       |
| <b>Step 2: Install as PWA App</b>     | In Chrome or Edge, click the <b>Install LinkForge</b> icon in the address bar (looks like a monitor with a down arrow). On iPad/iPhone, tap <i>Share</i> ➔ <i>Add to Home Screen</i> .                                                             |
| <b>Step 3: Install Extension</b>      | On his desktop browser, visit <b>/admin/extensions</b> . For Firefox: click <b>1-Click Direct Install (Firefox XPI)</b> . For Chrome/Edge: unzip dashforge-chrome.zip and Load Unpacked in <a href="chrome://extensions">chrome://extensions</a> . |
| <b>Step 4: Pin Favorite Sites</b>     | On the Home page, click <b>Add Pinned Shortcut</b> to pin his daily sites (BBC News, weather, sports, banking).                                                                                                                                    |
| <b>Step 5: Book Vault &amp; EPUBs</b> | Show him the <b>Book Vault</b> page. Search any author (e.g. <i>Griff Hosker</i> , <i>Bernard Cornwell</i> ), click <b>Auto-Grab Book</b> , and show him 1-click EPUB downloads for his e-reader or tablet.                                        |

### 5. Remote VPS Deployment, Cloudflare Tunnel & Live Updates

#### Cloudflare Tunnel Configuration (linkforge.okapitek.uk):

In your Cloudflare Zero Trust / Tunnel dashboard, add a Public Hostname rule:

- **Subdomain:** linkforge • **Domain:** okapitek.uk
- **Type:** HTTP • **URL:** localhost:5006 (or linkforge:5000 if sharing Docker bridge)
- **Note:** LinkForge includes ProxyFix middleware, ensuring HTTPS redirects, PWA Web Share Target, and client IPs operate with zero configuration over Cloudflare.

#### How Future Updates Work (Zero-Terminal / 100% In-Browser):

1. You push new code to GitHub from your PC (git push origin main).
2. Within seconds, your VPS and remote instances detect the new GitHub commit.
3. A glowing badge **UPDATE AVAILABLE** appears in the left sidebar.
4. In Settings, click **'Update LinkForge'** — LinkForge pulls the update and restarts in 2 seconds with zero data loss!

## 6. Firefox Extension (.XPI) Signing & In-App Distribution

Mozilla Firefox enforces strict cryptographic signatures on all add-ons. To distribute your LinkForge extension to users on standard release Firefox with zero security warnings and 1-click installation, sign the add-on through Mozilla's free self-distribution portal and save it to the LinkForge static downloads directory.

|                                    |                                                                                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1: Mozilla Portal</b>      | Go to <a href="https://addons.mozilla.org/developers/addon/submit/distribution">https://addons.mozilla.org/developers/addon/submit/distribution</a> and log in. |
| <b>Step 2: Self-Distribution</b>   | Choose ' <b>On your own</b> ' (signs the file for direct distribution without publishing to public store).                                                      |
| <b>Step 3: Upload Zip</b>          | Upload <code>X:\.config\docker\linkforge\static\downloads\dashforge-firefox.zip</code> .                                                                        |
| <b>Step 4: Download Signed XPI</b> | Mozilla's validator verifies the manifest (0 errors) and signs it in ~1-2 min. Download the signed .xpi.                                                        |
| <b>Step 5: Place in LinkForge</b>  | Save the file to replace: <code>X:\.config\docker\linkforge\static\downloads\dashforge-firefox.xpi</code> .                                                     |
| <b>Step 6: 1-Click Install</b>     | Now, visiting <code>/admin/extensions</code> allows anyone on Firefox to 1-click install with zero warning screens!                                             |

## 7. Cleaning Test Data (The Nuke Data Tab)

To clean any restored test bookmarks on a fresh container while leaving all AI configurations intact:

1. Open Admin Panel → **Nuke Data** (`/admin/nuke`).
2. Click '**Preset: Clean Restored Data (Keep AI Settings)**'.
3. Type **NUKE** in the confirmation box and click **Confirm Nuke**.
4. All test links, videos, and articles are wiped clean, while the Smart Master Switch, YouTube fetchers, and AI routing remain 100% saved.

## 8. Troubleshooting & Maintenance Commands Cheat Sheet

| Action / Problem                 | Command to Run in PowerShell / Bash                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------|
| Restart LinkForge Container      | <code>docker compose restart</code>                                                                  |
| Rebuild from scratch (forced)    | <code>docker compose up -d --build --force-recreate</code>                                           |
| View live application logs       | <code>docker logs -f linkforge</code>                                                                |
| Discard local unpushed changes   | <code>git reset --hard origin/main</code>                                                            |
| Repackage browser extensions     | <code>python package_extensions.py</code>                                                            |
| Create manual full system backup | <code>python -c "from services.backup import create_auto_backup; print(create_auto_backup())"</code> |